Analysing a dataset with SQL

Moneyball Challenge

Project Overview and Objectives

In this challenge, you will use SQL to analyse a sports dataset. Much like the "Moneyball" scenario where baseball data was analysed to make better player decisions, you'll use SQL queries to extract insights from a dataset of football players, teams, and their performance statistics. Your goal is to help a football team's management find patterns that can be used to make strategic decisions.

Dataset Description

The dataset comprises three main tables: Players, Teams, and Transfers. Each table contains specific information relevant to its entity, as detailed below.

Players Table

player_id	name	position	team_id	goals_scored	assists	matches_played	salary
1	Lionel Messi	Forward	1	30	10	25	50000000
2	Cristiano Ronaldo	Forward	2	28	7	23	45000000
3	Kylian Mbappe	Forward	3	27	5	22	30000000
4	Neymar Jr	Forward	4	20	10	20	35000000
5	Kevin De Bruyne	Midfielder	3	10	15	24	20000000
6	Robert Lewandowski	Forward	2	35	3	26	45000000
7	Mohamed Salah	Forward	3	25	8	22	25000000
8	Erling Haaland	Forward	4	40	4	25	30000000
9	Virgil van Dijk	Defender	4	5	1	25	15000000
10	Sergio Ramos	Defender	1	3	2	21	10000000

Teams Table

team_id	team_name	wins	losses	draws	league_position	budget
1	FC Barcelona	20	3	2	3	600000000
2	Juventus	18	5	1	4	500000000
3	Paris Saint-Germain	22	2	3	1	700000000
4	Manchester City	21	4	1	2	650000000

Transfers Table

player_id	from_team_id	to_team_id	transfer_fee
5	1	3	80000000
9	2	4	70000000
10	3	1	90000000

SQL Task 1: Top Performers

Query: Write a SQL query to list the top 5 players who have the highest number of goals scored. Include their name, team, goals scored, and position.

Expected Output: A table showing player names, their team names, goals scored, and their positions.

SQL Query:

```sql
select p.name as player_name, t.team_name, p.goals_scored, p.position from players p
JOIN teams t on p.team_id = t.team_id order by goals_scored desc limit 5
```

Output:

player_name	team_name	goals_scored	position
Erling Haaland	Manchester City	40	Forward
Robert Lewandowski	Juventus	35	Forward
Lionel Messi	FC Barcelona	30	Forward
Cristiano Ronaldo	Juventus	28	Forward
Kylian Mbappe	Paris Saint-Germain	27	Forward

Explanation:

- Joins players and teams tables.
- Orders by goals_scored to get top scorers.
- Returns top 5 highest-performing players.

SQL Task 2: Best Value for Money Players

Query: Write a SQL query to identify the top 3 players who provide the best value for money. "Value for money" is defined as the number of goals scored per unit of salary (goals/salary). Show player name, team name, goals, salary, and their calculated value-for-money ratio.

Expected Output: A table with player names, their team names, goals, salaries, and their value-for-money ratio, sorted in descending order.

SQL Query:

```
select p.name as player_name, t.team_name, p.goals_scored, p.salary,
ROUND((p.goals_scored * 1000000.0) / p.salary, 1) AS value_for_money_ratio
FROM players p
JOIN teams t
ON p.team_id = t.team_id
ORDER BY value_for_money_ratio DESC
LIMIT 3;
```

Output:

player_name	team_name	goals_scored	salary	value_for_money_ratio
Erling Haaland	Manchester City	40	30000000	1.3
Mohamed Salah	Paris Saint-Germain	25	25000000	1.0
Kylian Mbappe	Paris Saint-Germain	27	30000000	0.9

Explanation:

- Calculates goals-to-salary ratio for each player.
- Scales and rounds the ratio for readability.
- Returns top 3 most cost-effective players.

SQL Task 3: Team Performance Summary

Query: Write a SQL query to generate a summary report of each team's performance. The report should show the team name, number of wins, losses, draws, total goals scored by all its players, and the team's league position.

Expected Output: A table listing team names, their number of wins, losses, draws, total goals, and league position.

SQL Query:

```
SELECT
t.team_name, t.wins, t.losses, t.draws, SUM(p.goals_scored) AS total_goals, t.league_position
FROM teams t LEFT JOIN players ON t.team_id = p.team_id
GROUP BY
t.team_id, t.team_name, t.wins, t.losses, t.draws, t.league_position
ORDER BY
t.league_position;
```

Output:

team_name	wins	losses	draws	total_goals	league_position
Paris Saint-Germain	22	2	3	62	1
Manchester City	21	4	1	65	2
FC Barcelona	20	3	2	33	3
Juventus	18	5	1	63	4

Explanation:

- Joins teams with players data.
- Aggregates total goals scored per team.
- Displays overall team performance metrics.

SQL Task 4: Player Transfer Analysis

Query: Write a SQL query to find all players who have been transferred to a different team. Include player name, original team name, new team name, and the transfer fee.

Expected Output: A table showing player names, their original team, the team they transferred to, and the transfer fee.

SQL Query:

```
SELECT  p.name AS player_name,
t_from.team_name AS original_team,
t_to.team_name AS new_team,
tr.transfer_fee
FROM transfers tr
JOIN players p ON tr.player_id = p.player_id
JOIN teams t_from ON tr.from_team_id = t_from.team_id
JOIN teams t_to ON tr.to_team_id = t_to.team_id;
```

Output:

player_name	original_team	new_team	transfer_fee
Kevin De Bruyne	FC Barcelona	Paris Saint-Germain	80000000
Sergio Ramos	Paris Saint-Germain	FC Barcelona	90000000
Virgil van Dijk	Juventus	Manchester City	70000000

Explanation:

- Joins transfers with player and team details.
- Displays original team, new team, and transfer fee.
- Helps analyse player movement between teams.

SQL Task 5: Identify Potential Underperformers

Query: Write a SQL query to identify players who have played more than 10 matches but have scored fewer than 3 goals. Include their name, team, matches played, and goals scored.

Expected Output: A table with player names, their team, number of matches played, and their goals scored.

SQL Query:

```
select p.name as player_name,  t.team_name, p.matches_played, p.goals_scored
from players p join teams t on p.team_id = t.team_id
where p.matches_played > 10 and p.goals_scored < 4
```

Output:

player_name	team_name	matches_played	goals_scored
Sergio Ramos	FC Barcelona	21	3

Explanation:

- Filters players based on matches played and goals scored.
- Identifies players with low output despite high playtime.
- Helps flag potential underperformers.

SQL Task 6: Team Budget Efficiency

Query: Write a SQL query to find the top 3 teams that have achieved the highest number of wins with the lowest budget. Show the team name, number of wins, and their budget.

Expected Output: A table with team names, number of wins, and their budget, sorted by the best win-to-budget ratio.

SQL Query:

```
select team_name, wins, budget,ROUND((wins * 1000000.0)/budget,2) as win_to_budget_ratio
FROM teams ORDER BY win_to_budget_ratio DESC
LIMIT 3;
```

Output:

team_name	wins	budget	win_to_budget_ratio
Juventus	18	500000000	0.04
FC Barcelona	20	600000000	0.03
Paris Saint-Germain	22	700000000	0.03

Explanation:

- Compares team wins relative to budget.
- Calculates win-to-budget efficiency ratio.
- Returns top 3 most budget-efficient teams.